

Bank Management System

Developed by Jayesh Gaikwad HTML • CSS • JavaScript • Python • Flask • MongoDB GitHub + Render Deployment

1. Introduction

A web-based banking administration project for managing customers, accounts, transactions and loans. Flask handles backend routes and MongoDB stores flexible document records.

2. Objectives

Centralize banking records, provide reusable CRUD-ready modules, demonstrate Python full-stack skills, use MongoDB and create a GitHub/Render deployment structure.

3. Technology Stack

Frontend: HTML5, CSS3, JavaScript. Backend: Python Flask. Database: MongoDB/MongoDB Atlas. Version control: GitHub. Production server: Gunicorn. Hosting: Render.

4. Modules

Core modules include dashboard, customers, accounts, transactions, loans, employees, cards, reports and settings. Extended modules include KYC, transfers, deposits, statements, security, analytics and support.

5. Database Design

Collections include customers, accounts, transactions, loans, employees and cards. MongoDB documents can be indexed and validated as the project grows.

6. Application Workflow

The browser sends requests to Flask. Flask validates form data and communicates with MongoDB through PyMongo. Jinja templates render the response while CSS and JavaScript provide the interface.

7. Security

Production use should add authentication, role-based authorization, HTTPS, CSRF protection, validation, secure password hashing, rate limiting and audit logging. Do not use real banking data in a student demo.

8. Deployment

Push the source to GitHub, connect the repository to Render, install dependencies and start Gunicorn. Configure MONGO_URI and SECRET_KEY as Render environment variables.

9. Testing & Future Scope

Test routes, forms, database connectivity, responsive layout and deployment. Future improvements include login, roles, PDF statements, charts, balance rules, automated tests and richer analytics.

10. Conclusion

The project demonstrates practical full-stack development using Flask and MongoDB and provides a portfolio foundation for Python Full Stack or Backend Developer roles.